

Data assembled

Three patient CD4⁺ atlases downloaded, harmonised to approved HGNC symbols, and integrated into one single-cell object (**no pseudobulk**, raw counts preserved). A genome-scale CRISPRi perturbation atlas provides the reversal substrate.

Dataset	Role	Scale
Atlas B — Nat Med 2025 inflammation	Disease direction (source 1)	1,781,064 CD4 cells · 20 diseases · 960 donors
Atlas C — Yasumizu circulating CD4	Disease direction (source 2)	103,153 CD4 cells · MS / MG / SLE / HC · 13 donors
Integrated B+C object	Stage-1 input	1,884,217 cells × 17,165 genes
Perturb-seq (Marson 2025)	Reversal substrate (Stage 2)	~18k gene KDs × 3 contexts (DE tier, 16.8 GB)
Open Targets + eQTL Catalogue	Genetic evidence layer	239 MS genetic targets · CD4 eQTL panel (11 datasets queried)

Every extra candidate dataset was overlap-checked against the atlases before download, so no patient is double-counted — critical because the atlases are themselves meta-analyses of public data, and a re-download would break the replication filter.

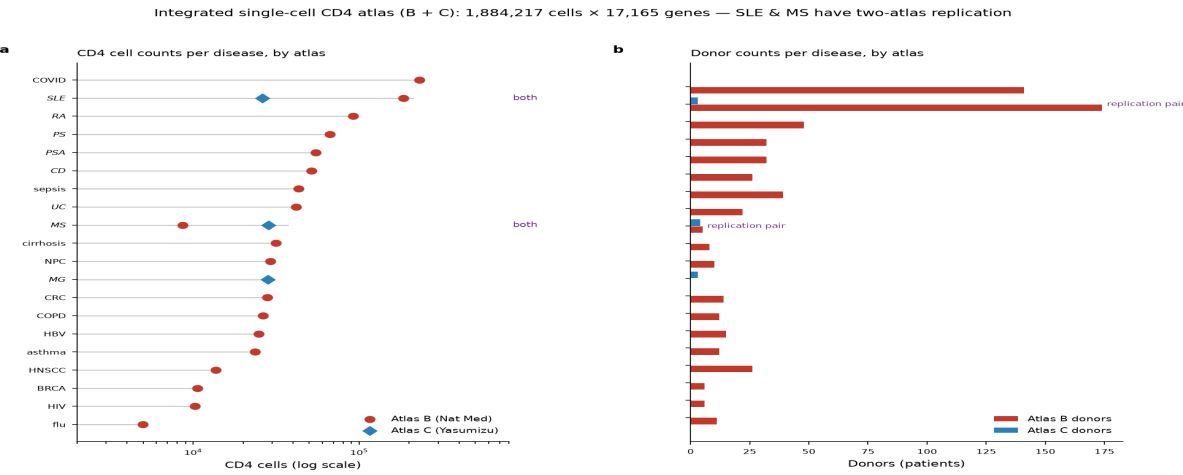


Fig 1. Integrated B+C CD4 atlas coverage. SLE and MS are the only autoimmune diseases present in both atlases (replication pairs). MS: 8,652 cells / 5 donors in B; 28,340 / 4 in C.

Genetic evidence layer (eQTL / GWAS integration)

To move Stage 1/2 from correlational to **causal**, each gene is annotated with human-genetics support for MS: the Open Targets genetic-association score, the max **locus-to-gene (L2G)** score across MS GWAS credible sets, and whether the gene carries a **cis-eQTL in CD4⁺ T cells**. We queried an 11-dataset CD4 panel from the eQTL Catalogue (OneK1K single-cell CD4 subsets, Schmiedel CD4/Th1/Th2/Th17/Treg, BLUEPRINT, Perez); 10 of those datasets returned at least one hit. The result is a genetic-priority weight for reversal scoring and node prioritisation.

239	231	83	73
MS genetic targets	with credible-set L2G	with a CD4 cis-eQTL	high-confidence (L2G>0.5 & CD4 eQTL)

The 73-gene high-confidence set (L2G>0.5 *and* a CD4 cis-eQTL) is led by **CD40**, **TNFRSF1A**, **BACH2**, **TRAF3**, **SH2B3**, **FCRL3**, **EVI5** and **AHI1** — canonical, CD4-relevant MS genes. Several strong genetic anchors score high on L2G but did *not* show a significant eQTL in this CD4 panel — notably **IL7R**, **JAK1**, **CD58** and **TNFSF14** (genetic-only tier); JAK1 remains of interest as an approved-JAK-inhibitor target, and IL2RA/CD58 are established MS genes worth carrying forward on genetic grounds despite the panel miss.

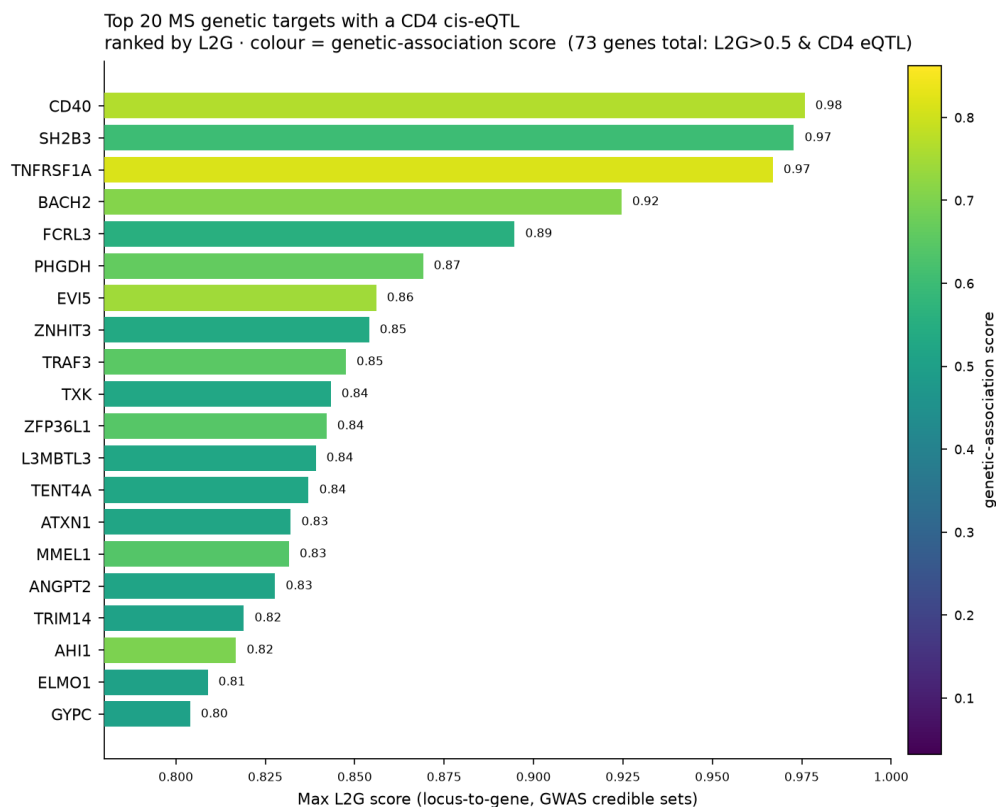


Fig 2. Top 20 MS genes with a CD4 cis-eQTL, ranked by L2G. Bar = L2G; colour = genetic-association score.